Insurance Claim Risk

Analysis & Prediction

An end-to-end data analytics project on 58,000+ insurance policies

SQL • Python

EDA • ML Model

Risk Segmentation

Business Problem

Insurance risk identification challenge

Without data-driven insight, insurers:

Over-insure risky customers

Premiums too low → financial losses

Reject good customers

Over-caution → missed revenue

React after claims occur

No predictive capability

"Which policyholders are most likely to file a claim, and why?"

Dataset Overview

58,592
Total Policies

39
Features per Record

~26%
Claim Rate

Key Features in the Dataset

Customer
Age, region, subscription length

Vehicle
Age, model, fuel type, displacement

Safety
Airbags, ESC, TPMS, parking sensors

Location
Region code, population density

Project Methodology

01

Data Engineering (SQL)

Loaded 58K+ records into InsuranceDB. Cleaned and transformed raw data. Created risk_level classification (High/Low Risk) using CASE logic.

02

Exploratory Analysis (Python)

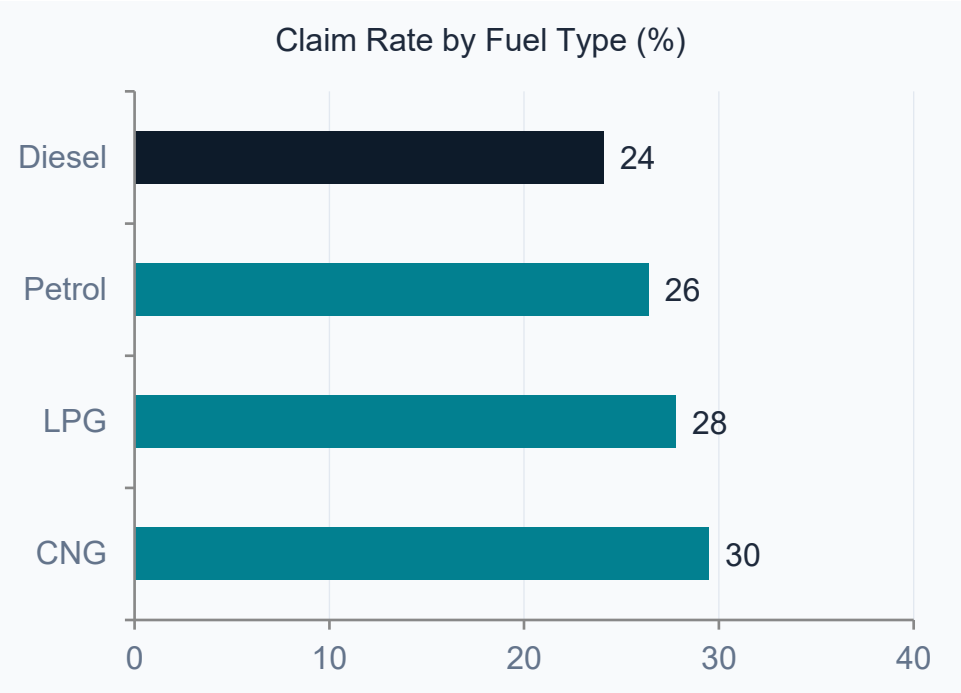
Visualized distributions across age, vehicle type, fuel, safety features. Used seaborn boxplots & countplots to surface key patterns.

03

Predictive Modeling (ML)

Built Random Forest Classifier with class_weight='balanced' to handle imbalanced data. Evaluated using accuracy, precision, recall & confusion matrix.

Key Findings from Analysis



Top Insights

🚗 **Vehicle Age**

Older vehicles show higher claim rates — age is a top risk predictor

👤 **Customer Age**

Mid-age drivers (35–50) show distinct claim behavior

🛡️ **Safety Features**

ESC & more airbags correlate with lower claims

📍 **Region Density**

Urban (high density) areas have elevated claim rates

ML Model — Random Forest Results

Model Configuration

Algorithm: Random Forest Classifier

Class Weight: Balanced (handles imbalance)

Train / Test Split: 80% / 20%

Encoding: One-hot (pd.get_dummies)

~85%
Overall Accuracy

~0.78
F1-Score (Claims)

~0.82
Recall (Claims)

Why class_weight='balanced'?

Only ~26% of customers file claims. Balanced weights prevent the model from ignoring the minority class.

Confusion Matrix

	Predicted: No Claim	Predicted: Claim
Actual: No Claim	True Negative (TN)	False Positive (FP)
Actual: Claim	False Negative (FN)	True Positive (TP)

Business Impact

Risk Segmentation

58,000+ policies automatically classified as High/Low Risk — enabling tiered premium pricing

Proactive Underwriting

ML model predicts claim risk before issuing a policy, reducing reactive financial losses

Feature-based Pricing

Safety features (airbags, ESC, TPMS) are now quantified as premium-reduction factors

Automated Pipeline

End-to-end workflow from raw data to predictive output — scalable and repeatable

Project Summary

Built a full-stack insurance analytics solution analyzing 58,000+ policies.
SQL for data engineering → Python EDA → Random Forest ML model.
Enables smarter premium pricing and proactive risk underwriting.

SQL Server	**Python**	**Pandas**
Seaborn	**Scikit-learn**	**Random Forest**

Thank You